

Research Statement

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My research lies at the interface of analysis, geometry, topology, and mathematical physics. A central question guiding my work is: *how can techniques from global analysis and index theory be used to give rigorous constructions of geometric/topological invariants arising in mathematical physics?* A recurring theme is localization. Witten deformation concentrates spectral information near critical sets, heat-kernel analysis connects local geometry with global spectral invariants, and compactifications of configuration spaces provide a geometric framework for analyzing the singularities in Feynman graph integrals. My research develops along three interconnected directions:

- (a) Feynman graph integrals in holomorphic quantum field theory and their applications to higher-genus B-model invariants;
- (b) analytic torsion and higher Cheeger–Müller theory, using Witten deformation to compare analytic and topological invariants;
- (c) spectral theory of Schrödinger-type operators on noncompact manifolds, motivated in part by Landau–Ginzburg models.

These directions come together in my long-term program toward a higher-genus Calabi–Yau/Landau–Ginzburg correspondence on the B-model side.

1 Overview of my results

My main results to date can be summarized as follows.

- **Feynman graph integrals and holomorphic field theory.** With Minghao Wang, we proved the convergence of Feynman graph integrals on closed real-analytic Kähler manifolds in the sense of Cauchy principal values [20]. On Calabi–Yau threefolds, this gives a direct and rigorous realization of the candidate higher-genus B-model invariants proposed by Bershadsky–Cecotti–Ooguri–Vafa. We subsequently proved that three natural rigorous renormalization procedures give the same answer and derived a gauge anomaly formula [21]. With Xinxing Tang, we also proved the CY/LG correspondence for Weil–Petersson metrics and tt^* -structures, giving a genus-zero comparison on the B-model side [19].
- **Analytic torsion and higher torsion.** Using Witten deformation by specially chosen non-Morse functions, I developed a localization method adapted to decompositions of manifolds rather than isolated Morse critical points. This gives a new purely analytic proof of the general gluing formula for analytic torsion and, for families, a substantially simpler proof of the gluing formula for Bismut–Lott torsion forms [22, 23]. With Martin Puchol, we proved a version of the higher Cheeger–Müller/Bismut–Zhang theorem using generalized Morse functions and resolved the main analytic difficulty caused by birth–death singularities in families [17].
- **Spectral theory on noncompact manifolds.** With Xianzhe Dai, we proved exponential decay of eigenforms and identified the analytic Witten complex with a finite-dimensional complex generated by critical points on noncompact manifolds. We then developed heat-kernel expansions and a local index theorem for Callias-type operators [9, 10]. In later work we established classical and semiclassical Weyl laws [11]. With Maxim Braverman and Dai, we obtained an intrinsic criterion for the classical Weyl law on general complete manifolds and constructed examples

showing that the law can fail when the criterion is violated [8]. More recently, I applied these ideas to show that the weighted Laplacian on every complete gradient Ricci shrinker satisfies the classical Weyl law [24].

2 Feynman graph integrals and holomorphic field theory

In perturbative quantum field theory, Feynman graph integrals encode the contributions associated with interactions of particles. In mathematics, they also provide a striking bridge between physical theories and geometric or topological invariants. In topological quantum field theory (TQFT), such integrals converge in the Lebesgue sense and give rise to important topological invariants [1, 7, 16].

In contrast, in general QFTs, Feynman graph integrals develop singularities, so they are typically not well defined without *renormalization*; moreover, the renormalization procedures are often noncanonical. Our work asks whether, for holomorphic field theories on Kähler manifolds, configuration-space compactifications can be used to analyze the singularities and thereby define the resulting Feynman graph integrals canonically.

More specifically, let Γ be a graph with vertex set Γ_0 and edge set Γ_1 . Assigning a point $x_v \in M$ to each vertex $v \in \Gamma_0$ gives

$$x = (x_v)_{v \in \Gamma_0} \in M^{\Gamma_0} := \prod_{v \in \Gamma_0} M.$$

Each edge $e = (v, w)$ contributes a propagator $P_e(x_v, x_w)$, typically constructed from the Green kernel of an elliptic operator and possibly involving its derivatives. Schematically, the graph weight and its associated integral are

$$\widetilde{W}_\Gamma(x) = \prod_{e=(v,w) \in \Gamma_1} P_e(x_v, x_w), \quad W_\Gamma = \int_{M^{\Gamma_0}} \widetilde{W}_\Gamma.$$

The propagator is singular at the diagonal. Thus $\widetilde{W}_\Gamma$ is smooth only on the configuration space

$$\text{Conf}_\Gamma(M) = \left\{ (x_v)_{v \in \Gamma_0} \in M^{\Gamma_0} : x_v \neq x_w \text{ for every } e = (v, w) \in \Gamma_1 \right\},$$

and the integral over M^{Γ_0} may diverge near the collision loci (diagonals). The main difficulty occurs when several vertices collide simultaneously, producing nested singularities.

Holomorphic QFTs exhibit sufficiently nice geometric structures to suggest that graph integrals in this setting may admit canonical renormalizations. In joint work with Wang, we proved the convergence of these graph integrals on closed real-analytic Kähler manifolds [20]. Using heat-kernel techniques together with Getzler rescaling from local index theory, we show that the graph integrands extend canonically to the Fulton–MacPherson compactification of configuration spaces as forms with controlled singularities and therefore admit well-defined Cauchy principal-value regularizations.

When specialized to Calabi–Yau threefolds, this construction gives a mathematically rigorous definition of candidate higher-genus ($g \geq 2$) B-model invariants through the Feynman graph integrals proposed by BCOV in [2]. Thus it provides a direct analytic construction of the higher-genus closed-string B-model quantities predicted by physics.

A further question is whether different natural renormalization procedures produce the same invariant. With Wang, we proved that heat-kernel renormalization, zeta regularization, and Cauchy principal value regularization all give the same value for holomorphic field theories on closed real-analytic Kähler manifolds [21]. The difficulty is that these procedures organize the same singularities in very different ways: heat-kernel renormalization cuts off short time, zeta regularization uses meromorphic continuation, and principal value uses the geometry of the resolved diagonals. Wonderful compactifications in algebraic geometry provide a common geometric framework for analyzing the nested singularity loci underlying all three regularizations, allowing the three procedures to be compared directly. We also derived a gauge anomaly formula.

We will next extend the construction to families and study anomaly formulas for Feynman graph integrals under variations of Kähler and complex structures. The goal is to understand the differential forms on moduli spaces of Calabi–Yau manifolds obtained from fiberwise Feynman graph integrals.

Their construction should be analogous to that of the analytic torsion forms discussed in § 3. A second direction is field theory on manifolds with boundary, where my Witten-deformation gluing methods should separate bulk and boundary contributions.

3 Higher Cheeger–Müller/Bismut–Zhang theory

A classical problem in global analysis is to understand why spectral invariants of differential operators can coincide with invariants defined by topological or combinatorial methods. Ray–Singer analytic torsion is a central example: it is defined from regularized determinants of Hodge Laplacians, whereas Reidemeister torsion is constructed from a finite-dimensional cellular complex.

The defining data for Ray–Singer analytic torsion consist of a closed Riemannian manifold (M, g^{TM}) , a flat complex vector bundle $(F, \nabla^F) \rightarrow M$, and a Hermitian metric g^F on F . The analytic torsion is defined using the regularized determinants of the F -twisted Hodge Laplacians. Unlike the index, torsion depends on the full nonzero spectrum and is therefore a finer invariant. The Cheeger–Müller theorem identifies analytic and Reidemeister torsion, and Bismut–Zhang extended the comparison to general flat bundles using Witten deformation [6].

For families of manifolds, the corresponding problem is to compare differential-form-valued analytic torsion with higher topological torsion over the parameter space. Let $\pi : M \rightarrow S$ be a smooth fibration with fibers $Z_s = \pi^{-1}(s)$. Bismut and Lott generalized the numerical analytic torsion of a single manifold to an even differential form

$$\mathcal{T}_{\text{BL}}(M/S, F)$$

on the parameter space S [5]. Its degree-zero component recovers ordinary analytic torsion. On the topological side, higher Franz–Reidemeister, or Igusa–Klein, torsion can detect exotic smooth structures that are invisible to ordinary characteristic classes [12–14].

A higher Cheeger–Müller/Bismut–Zhang theorem should compare these analytic and topological classes. The difficulty is that earlier approaches remove different restrictions. The analytic method of Bismut–Goette [3] treats general flat bundles but requires the existence of a fiberwise Morse function satisfying fiberwise Thom–Smale transversality. On the other hand, the axiomatization approach in [18] avoids the fiberwise Morse-function existence assumption but works only for trivial flat bundles.

With Martin Puchol, we combine the advantages of these approaches. The axiomatization argument reduces the problem to the comparison of relative higher torsions. On the analytic side, the relative torsion is

$$\mathcal{T}_{\text{BL}}^{\text{rel}}(M/S, F) := \mathcal{T}_{\text{BL}}(M/S, F) - \mathcal{T}_{\text{BL}}(M/S, \underline{\mathbb{C}}^{\text{rk}(F)}),$$

where $\underline{\mathbb{C}}^{\text{rk}(F)}$ is the trivial flat bundle of the same rank as F ; the topological torsion is normalized in the same way.

We extend the analytic method of Bismut–Goette from fiberwise Morse functions to fiberwise generalized Morse functions, which always exist but are allowed to have birth–death singularities. This simultaneously handles general unitary flat bundles and removes the fiberwise Morse-function existence restriction, yielding the expected higher Cheeger–Müller/Bismut–Zhang comparison for the relative torsions, while still assuming Thom–Smale transversality [17].

The main analytic difficulty is that the classical Witten-deformation estimates cease to be uniform near a birth–death singularity. At a birth–death point, two critical points merge and the Hessian develops a zero direction, so the standard quadratic Morse model degenerates. We resolve this difficulty by combining Bismut–Lebeau’s localization techniques [4] with the noncompact Agmon estimates from my earlier joint work with Dai [9].

Future direction: removing transversality by higher structures

The remaining step toward a fully general theorem is to remove global Thom–Smale transversality. Since a single fiberwise metric satisfying this condition need not exist globally, my approach is to choose Thom–Smale metrics locally on an open cover of the base and glue the resulting Morse complexes by higher continuation data.

One can cover the base by sufficiently small open sets

$$S = \bigcup_i U_i$$

such that, over each U_i , there exists a smooth family of metrics in the vertical direction satisfying the Thom–Smale condition. These local Thom–Smale complexes do not agree literally on overlaps. On a double intersection $U_i \cap U_j$, two choices of metric give a continuation map between the corresponding complexes. On a triple intersection $U_i \cap U_j \cap U_k$, the two possible compositions of continuation maps agree only up to chain homotopy. On higher intersections, one must introduce higher homotopies satisfying suitable coherence relations.

The project therefore has three steps. First, construct the local Thom–Smale complexes and their coherent higher transition data, including their behavior near birth–death loci. Second, use this higher structure to define the Čech relative topological torsion and prove its independence of auxiliary choices (with some normalization). Third, apply Witten deformation locally and prove that the resulting comparisons are compatible on multiple intersections, thereby comparing the Čech torsion with the Bismut–Lott analytic torsion form. The main difficulty is to control simultaneously the higher homotopy structures on overlaps and the singular analysis near birth–death points.

4 Analysis of Schrödinger-type operators on noncompact manifolds

A basic problem in the Landau–Ginzburg (LG) B-model is to define and study spectral invariants of Schrödinger-type operators on noncompact Kähler manifolds. My approach is to develop the spectral and heat-kernel theory of these operators directly in the noncompact setting, with the superpotential providing the confining potential.

More precisely, an LG model is specified by a noncompact Kähler manifold X and a holomorphic function f , called the superpotential. The associated operator on differential forms has the form

$$\Delta_f = \Delta + |\nabla f|^2 + \text{a zero-order term involving } \text{Hess}(f).$$

The genus-one invariant of the LG B-model is expected to be governed by a regularized determinant of Δ_f .

A major obstacle in this program is that many fundamental tools of spectral geometry rely heavily on compactness and do not directly extend to noncompact manifolds with potentials. In a series of joint works with X. Dai, we developed analytic foundations for Schrödinger-type operators on noncompact manifolds, which are needed to define higher-genus BCOV invariants in the LG B-model. We proved exponential decay of eigenfunctions near infinity (Agmon estimates), which leads to the well-definedness of Thom–Smale–Witten cohomology for Morse functions on noncompact manifolds [9]. We also derived heat-kernel asymptotic expansions for such operators [10], allowing us to extend the local index theorem to noncompact manifolds with potentials.

Weyl law for Schrödinger operators on noncompact manifolds

A basic question in spectral geometry is whether the classical Weyl law continues to hold for Schrödinger operators on noncompact manifolds, where both the geometry at infinity and the growth of the potential can affect the eigenvalue distribution.

Using the heat-kernel expansion developed in [10], we established in [11] a *non-semiclassical Weyl law* for Schrödinger operators on noncompact manifolds with bounded geometry and potentials satisfying a doubling condition. This is the first Weyl law in a general noncompact geometric setting; prior results required explicit models of the geometry at infinity, such as Euclidean, hyperbolic, or cylindrical ends. Even in special cases, such as hyperbolic manifolds with rapidly growing potentials studied in [15], which fall within the scope of our result, the analysis is already delicate.

In subsequent work with Braverman and Dai, we removed both bounded geometry and the doubling condition [8]. Instead of imposing separate uniform assumptions on the manifold and the potential, we introduced an intrinsic quantity $c_\delta(\lambda)$ that measures their interaction at energy λ . We proved the Weyl

law when $c_\delta(\lambda) \rightarrow 0$, and constructed examples in which the Weyl law fails when this balance breaks down. Thus the result identifies a mechanism governing Weyl asymptotics rather than another list of sufficient hypotheses. More recently, I applied these ideas to show that the weighted Laplacian on every complete gradient Ricci shrinker satisfies the classical Weyl law [24].

My next goals are to obtain remainder estimates. On compact manifolds, refined remainder estimate have been obtained using Fourier integral operator methods. Extending these techniques to Schrödinger-type operators on noncompact manifolds would lead to a deeper understanding of spectral distributions in such geometric and analytic settings.

5 Toward a higher-genus Calabi–Yau/Landau–Ginzburg correspondence on the B-side

My long-term goal is to establish a higher-genus CY/LG correspondence on the B-model side. The approach is to combine the two analytic frameworks developed above: Feynman graph integrals for the compact Calabi–Yau theory and Schrödinger/heat-kernel analysis for the noncompact LG theory.

To place this problem in context, mirror symmetry relates symplectic invariants of a Calabi–Yau manifold to complex-geometric invariants of its mirror. The CY/LG correspondence gives a second bridge: it relates a compact Calabi–Yau theory to a Landau–Ginzburg model defined on a noncompact Kähler manifold equipped with a superpotential. The theory is organized by genus.

At genus zero, the B-model side is governed by tt^* -geometry, which can be viewed as a generalized variation of Hodge structures. With Tang, we proved the CY/LG correspondence for Weil–Petersson metrics and tt^* -structures [19]. This gives a rigorous genus-zero identification and provides the foundation for the higher-genus program.

At higher genus, even defining the relevant B-model quantities rigorously is difficult. The work described above now provides two complementary ingredients. The graph-integral theory constructs the compact Calabi–Yau B-model invariants, while the noncompact heat-kernel and Agmon estimates control the Schrödinger-type operators of the LG model. The next objective is to combine them on a noncompact Kähler manifold with a superpotential.

This combination introduces a new analytic difficulty. On a compact Calabi–Yau manifold, the singularities of a Feynman graph integral arise from the diagonals (where several vertices collide). In the Landau–Ginzburg setting, the underlying Kähler manifold is noncompact, so one must additionally control near infinity. These two degenerations can occur simultaneously: several vertices may collide while moving toward infinity together. A rigorous construction therefore requires a compactification that resolves both the collision diagonals and the behavior at infinity, together with uniform estimates for the propagators near the resulting corner strata. Establishing such estimates would give a rigorous definition of higher-genus BCOV invariants on the LG B-side.

The comparison with the Calabi–Yau invariants would then become a precise mathematical question rather than a formal physical prediction.

This program illustrates the unifying role of global analysis in my research. Witten deformation controls infinity and critical loci; heat kernels connect local geometry with spectral invariants; and compactifications convert ultraviolet divergence into geometry on boundary strata. My long-term goal is to develop these ideas into a common analytic framework for invariants arising in index theory and quantum field theory.

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